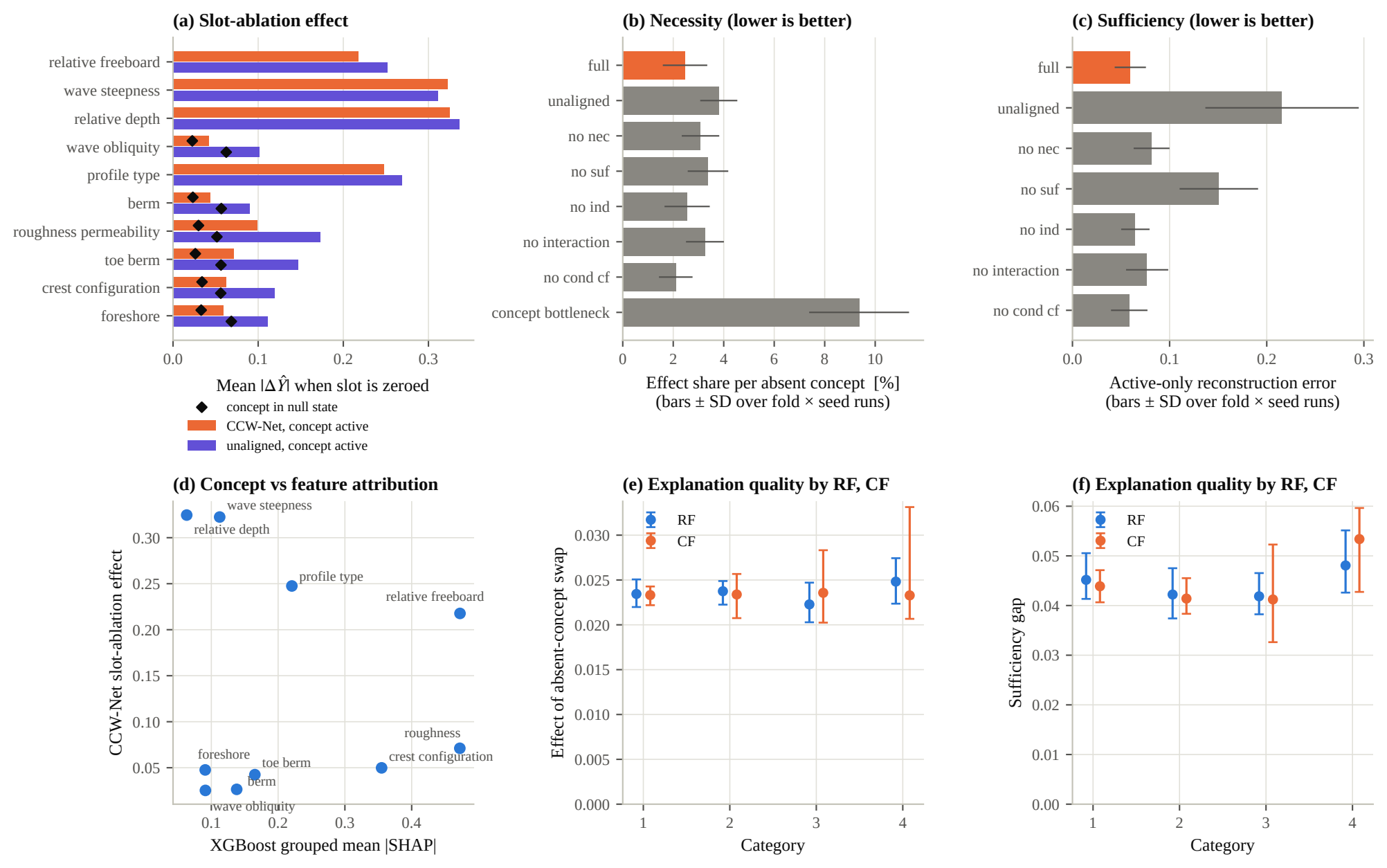
(a) Slot-ablation effect
relative freeboard
wave steepness
relative depth
wave obliquity
profile type
berm
roughness permeability
toe berm
crest configuration
foreshore
0.0
0.1
0.2
0.3
Mean |Δ Ŷ| when slot is zeroed
concept in null state
CCW-Net, concept active
unaligned, concept active
(b) Necessity (lower is better)
full
unaligned
no nec
no suf
no ind
no interaction
no cond cf
concept bottleneck
0
2
4
6
8
10
Effect share per absent concept [%]
(bars ± SD over fold × seed runs)
(c) Sufficiency (lower is better)
full
unaligned
no nec
no suf
no ind
no interaction
no cond cf
0.0
0.1
0.2
0.3
Active-only reconstruction error
(bars ± SD over fold × seed runs)
(d) Concept vs feature attribution
wave steepness
relative depth
profile type
relative freeboard
roughness
crest configuration
foreshore
toe berm
berm
wave obliquity
0.30
0.25
0.20
0.15
0.10
0.05
0.1
0.2
0.3
0.4
CCW-Net slot-ablation effect
XGBoost grouped mean |SHAP|
(e) Explanation quality by RF, CF
RF
CF
0.030
0.025
0.020
0.015
0.010
0.005
0.000
1
2
3
4
Effect of absent-concept swap
Category
(f) Explanation quality by RF, CF
RF
CF
0.06
0.05
0.04
0.03
0.02
0.01
0.00
1
2
3
4
Sufficiency gap
Category